

STAT 400 Final Report

I: Introduction

Initial Data Overview

This study uses the **NYC Flights dataset (2023)** from the `nycflights23` package. This is a package that contains detailed information on domestic flights departing from 3 New York City airports, including flight timing, delays, carrier information, aircraft characteristics, and weather conditions. This package is preloaded in R and requires no extra installation. 3 datasets are selected to make the analysis; they are flight, weather and planes. To get a better understanding of the dataset's content and how is it collected, please read the brief introduction below:

1. **Flight:** On-time data for all flights that successfully took off from NYC airports during 2023 recording various variables about the exact flight information such as departing airport of flight, scheduled departure time, total distance of flight, and tail number of plane. It should be noticed that this dataset only contains flights that have actual departure time during year 2023 instead of scheduled time, so this dataset also contains a few flights that are scheduled to leave in 2022. Same reason causes information about some seriously delayed flights that are scheduled to leave in 2023 not recorded but considering their number to the total dataset observations, they make little impact to comprehensiveness of data and accuracy of conclusion based on it.

	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	sched_arr_time	arr_delay	carrier	flight	tailnum	origin	dest	air_time	distance	hour	minute	time_hour	flight_date
1	2023	1	1	1	2038	203	328	3	205	UA	628	N25201	EWB	SMF	367	2500	20	38	2023-01-01 20:00:00	2023-01-01
2	2023	1	1	18	2300	78	228	135	53	DL	393	N830DN	JFK	ATL	108	760	23	0	2023-01-01 23:00:00	2023-01-01
3	2023	1	1	31	2344	47	500	426	34	B6	371	N807JB	JFK	BQN	190	1576	23	44	2023-01-01 23:00:00	2023-01-01
4	2023	1	1	33	2140	173	238	2352	166	B6	1053	N265JB	JFK	CHS	108	636	21	40	2023-01-01 21:00:00	2023-01-01
5	2023	1	1	36	2048	228	223	2252	211	UA	219	N17730	EWB	DTW	80	488	20	48	2023-01-01 20:00:00	2023-01-01
6	2023	1	1	503	500	3	808	815	-7	AA	499	N925AN	EWB	MIA	154	1085	5	0	2023-01-01 05:00:00	2023-01-01
7	2023	1	1	520	510	10	948	949	-1	B6	996	N2043J	JFK	BQN	192	1576	5	10	2023-01-01 05:00:00	2023-01-01
8	2023	1	1	524	530	-6	645	710	-25	AA	981	N918AN	EWB	ORD	119	719	5	30	2023-01-01 05:00:00	2023-01-01
9	2023	1	1	537	520	17	926	818	68	UA	206	N13113	EWB	IAH	258	1400	5	20	2023-01-01 05:00:00	2023-01-01

Above is the snapshot of the first few rows of the dataset.

This dataset is used as the primary dataset for merging because:

- This dataset is used as the primary dataset for merging because each row represents a single flight departing from one of the NYC airports in 2023, which

serves as the basic unit of airport operations and is the main focus for travelers, airlines, and airports.

- b. This dataset contains key identification variables that allow it to be merged with the other two datasets: time_hour and tailnum.

2. **Weather:** Hourly meteorological data from 3 NYC airports including time-related variables and weather-related quantitative variables, such as humidity and precipitation. Each row has its airport record and exact time in hours as identification.

	origin	year	month	day	hour	temp	dewp	humid	wind_dir	wind_speed	wind_gust	precip	pressure	visib	time_hour
1	JFK	2023	1	1	0	48.0	48.0	100.00	0	0.00000	0.000000	1e-02	1010.2	0.25	2023-01-01 00:00:00
2	JFK	2023	1	1	1	48.2	48.2	100.00	190	4.60312	5.297178	1e-02	1009.2	2.50	2023-01-01 01:00:00
3	JFK	2023	1	1	2	49.0	49.0	100.00	190	5.75390	6.621473	1e-04	1009.0	0.25	2023-01-01 02:00:00
4	JFK	2023	1	1	3	49.0	49.0	100.00	250	5.75390	6.621473	2e-02	1008.0	4.00	2023-01-01 03:00:00
5	JFK	2023	1	1	4	49.0	49.0	100.00	170	8.05546	9.270062	1e-04	1007.8	0.75	2023-01-01 04:00:00
6	JFK	2023	1	1	5	48.0	48.0	100.00	0	0.00000	0.000000	1e-04	1007.6	0.75	2023-01-01 05:00:00
7	JFK	2023	1	1	6	46.4	46.4	100.00	250	9.20624	10.594357	0e+00	1007.3	0.24	2023-01-01 06:00:00

Above is the snapshot of the first few rows of the dataset.

3. **Planes:** Plane metadata about plane-body-related variables for all plane tail numbers found in the FAA aircraft registry. *American Airways (AA) and Envoy Air (MQ) report fleet numbers rather than tail numbers so can't be matched.

	tailnum	year	type	manufacturer	model	engines	seats	speed	engine
1	N101DQ	2020	Fixed wing multi engine	AIRBUS	A321-211	2	199	0	Turbo-fan
2	N101DU	2018	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
3	N101HQ	2007	Fixed wing multi engine	EMBRAER-EMPRESA BRASILEIRA DE	ERJ 170-200 LR	2	80	0	Turbo-fan
4	N101NN	2013	Fixed wing multi engine	AIRBUS INDUSTRIE	A321-231	2	379	0	Turbo-fan
5	N102DN	2020	Fixed wing multi engine	AIRBUS	A321-211	2	199	0	Turbo-fan
6	N102DU	NA	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
7	N102HQ	2007	Fixed wing multi engine	EMBRAER-EMPRESA BRASILEIRA DE	ERJ 170-200 LR	2	80	0	Turbo-fan
8	N102NN	2013	Fixed wing multi engine	AIRBUS	A321-231	2	379	0	Turbo-fan
9	N102UW	1998	Fixed wing multi engine	AIRBUS INDUSTRIE	A320-214	2	182	0	Turbo-fan

Above are the first few rows of the dataset.

We will discuss more about the dataset, its variables, missing values, limitations, and how they are combined in section II.

Purpose and Meaning of Research

New York City is one of the most important transportation hubs in the eastern United States and serves as a major gateway for both domestic and international travel. Its airports—John F. Kennedy International Airport, LaGuardia Airport, and Newark

Liberty International Airport—handle enormous numbers of passengers every year. These airports connect travelers to cities across the United States and around the world, making New York a central entry point for tourism, business, immigration, and international trade. Because of the city's economic and cultural importance, the demand for flights in and out of the NYC area remains extremely high throughout the year.

This heavy traffic also makes delays a significant issue. When airports operate near capacity, even small disruptions such as weather conditions, air traffic congestion, or late arrivals can quickly affect many flights throughout the system. Delays in New York airports may not only impact local travelers but also create ripple effects across national and international flight networks. We believe exploring and understanding delay patterns is therefore important for both passengers and airlines. By identifying how delays vary by airport, season, or time of day, researchers and transportation planners can better understand congestion patterns and potentially improve scheduling, efficiency, and travel reliability.

The purpose of this study is to examine the factors that influence airline arrival delays and to understand how these effects vary across airlines. Flight delays are a persistent operational challenge in aviation, affecting airline efficiency, passenger experience, and airport congestion. This analysis focuses on how flight-level operational characteristics (such as departure delay, flight timing, and distance) and environmental conditions (such as weather) contribute to arrival delays. Additionally, the study investigates whether airlines differ in both their baseline delay levels and how strongly departure delays propagate into arrival delays. This is relevant for identifying systematic inefficiencies in air travel and for understanding the hierarchical structure of delay formation in airline operations.

Definition of Delay

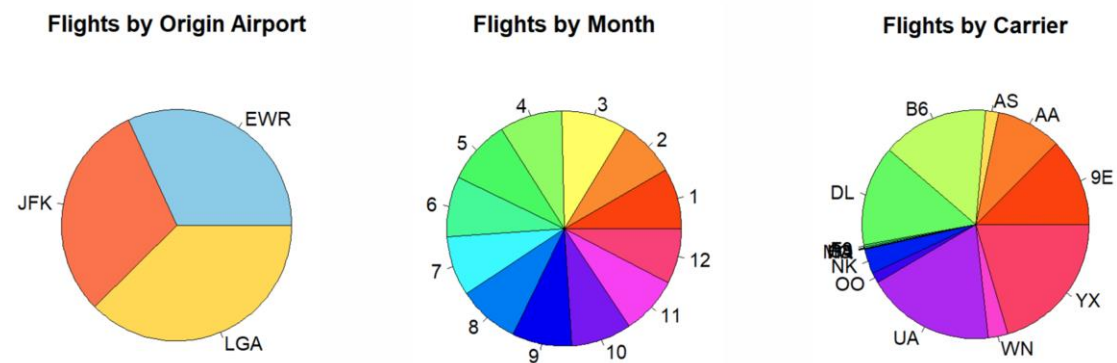
Because the data is about flights from U.S. airports, U.S. Department of Transportation's definition of delay will be adapted. Only flights arriving 15 minutes later than planned will be considered a delayed flight. Because there is no regulation of delayed departure, 15 minutes rule will be applied to departure, too. The delay

mentioned hereafter will follow this definition.

II: Explorative Data Analysis

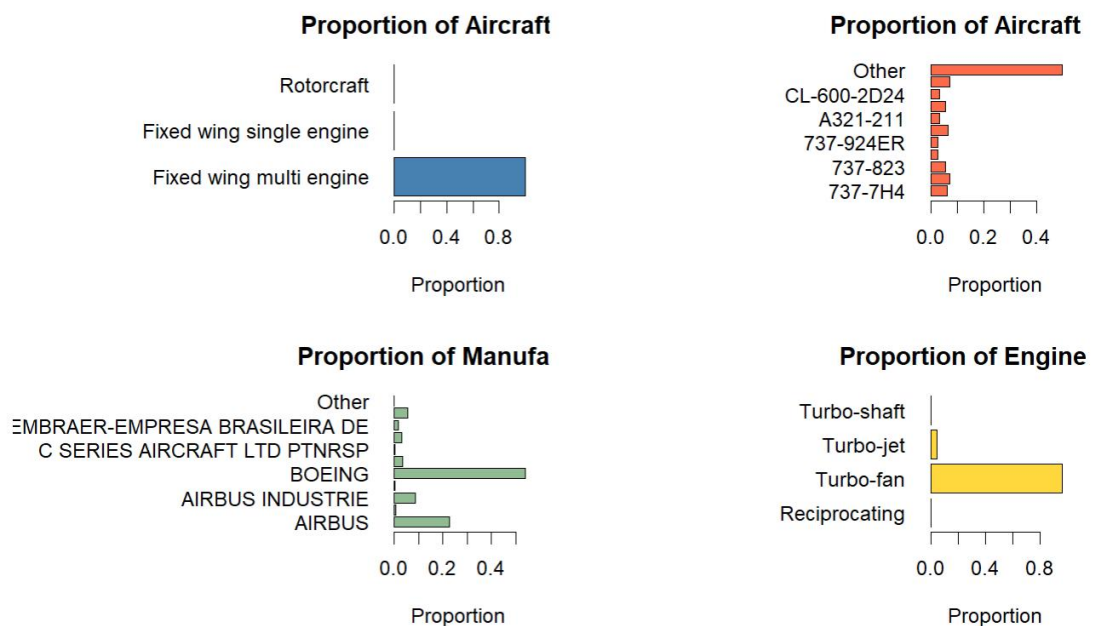
Some variables are not analyzed here because they don't contain valuable information, such as year (all identical values) and tailnum (all unique).

① pie graphs and bar graphs of valuable categorical variables



Origin Airport and Month: Data is evenly distributed for both variables.

Carrier: that flight activity is unevenly distributed across both airlines and scheduling patterns. A few carriers (such as YX, UA, and DL) account for a large share of flights, while others small airlines contribute relatively little combined.



Aircraft type: The aircraft type distribution shows that the overwhelming majority of

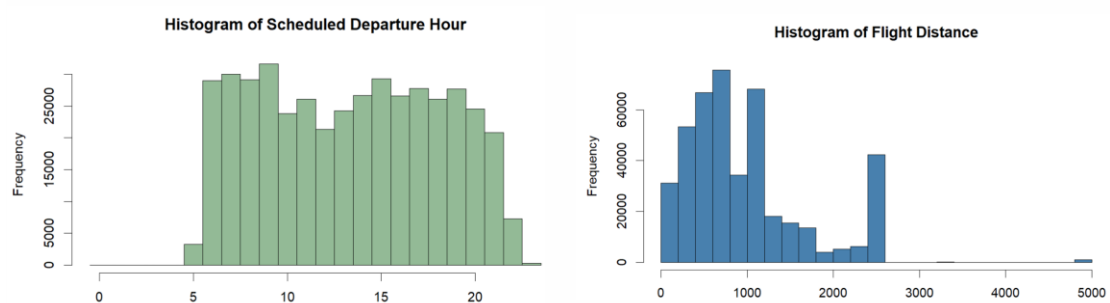
planes in the dataset are fixed-wing multi-engine aircraft. Fixed-wing single-engine aircraft and rotorcraft appear only rarely. This suggests that commercial passenger flights in the dataset are dominated by larger multi-engine aircraft designed for regular airline operations.

Aircraft model: The aircraft model distribution indicates that a small number of models account for a large proportion of the fleet, while many less common models are grouped into the “Other” category. Boeing 737 variants and several Airbus-related models appear frequently, showing that airlines rely heavily on a limited set of commonly used commercial aircraft models.

Manufacturer: The manufacturer's distribution is concentrated among a few major aircraft producers. Boeing represents the largest share of aircraft, followed by Airbus and Airbus Industrie, while other manufacturers contribute much smaller proportions. This reflects the dominance of Boeing and Airbus in the commercial aviation industry.

Engine Type: The engine type graph shows that turbo-fan engines overwhelmingly dominate the dataset. Turbo-jet, turbo-shaft, and reciprocating engines appear only in very small proportions. This is expected because modern commercial passenger aircraft primarily use turbo-fan engines due to their efficiency and suitability for long-distance air travel.

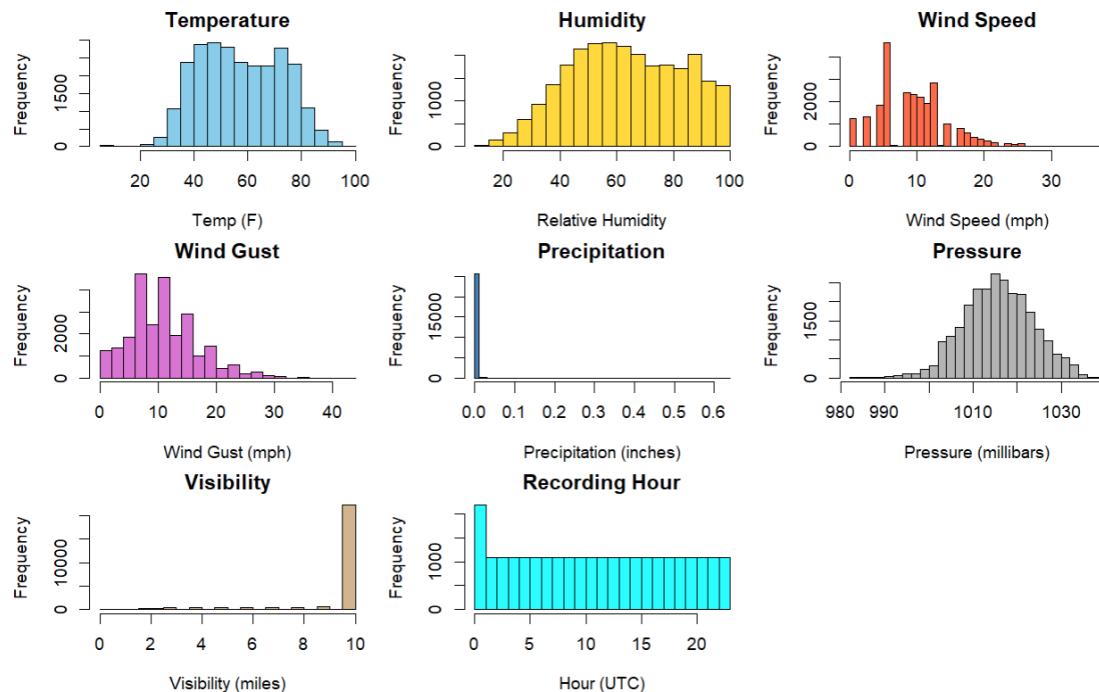
② Histograms of valuable quantitative variables



Scheduled departures: The scheduled departure time distributes pretty evenly during daytime hours, with fewer flights late at night or very early in the morning, reflecting typical travel demand patterns.

Flight distance: the distribution is right-skewed, with most flights being short to

medium range and only a small number of long-distance flights. Overall, these patterns suggest that both carrier dominance and operational scheduling play important roles in shaping flight activity.



Temperature: The temperature distribution appears roughly unimodal and moderately symmetric, with most observations concentrated between about 40°F and 80°F. Extremely low and extremely high temperatures are relatively uncommon. This suggests that most flights occurred under moderate weather conditions rather than severe temperature extremes.

Humidity: Its values are spread across a broad range but are concentrated mainly between 40% and 90%. The distribution suggests that moderately humid conditions were common throughout the dataset. Very low humidity values appear relatively rare compared to moderate and high humidity conditions.

Wind speed: the distribution is right-skewed, with most observations occurring at lower wind speeds between about 5 and 15 mph. Strong wind conditions are less common and appear as a long right tail. This indicates that severe wind conditions occurred infrequently relative to normal operating conditions.

Wind gust: Wind gust values also show a right-skewed distribution similar to wind

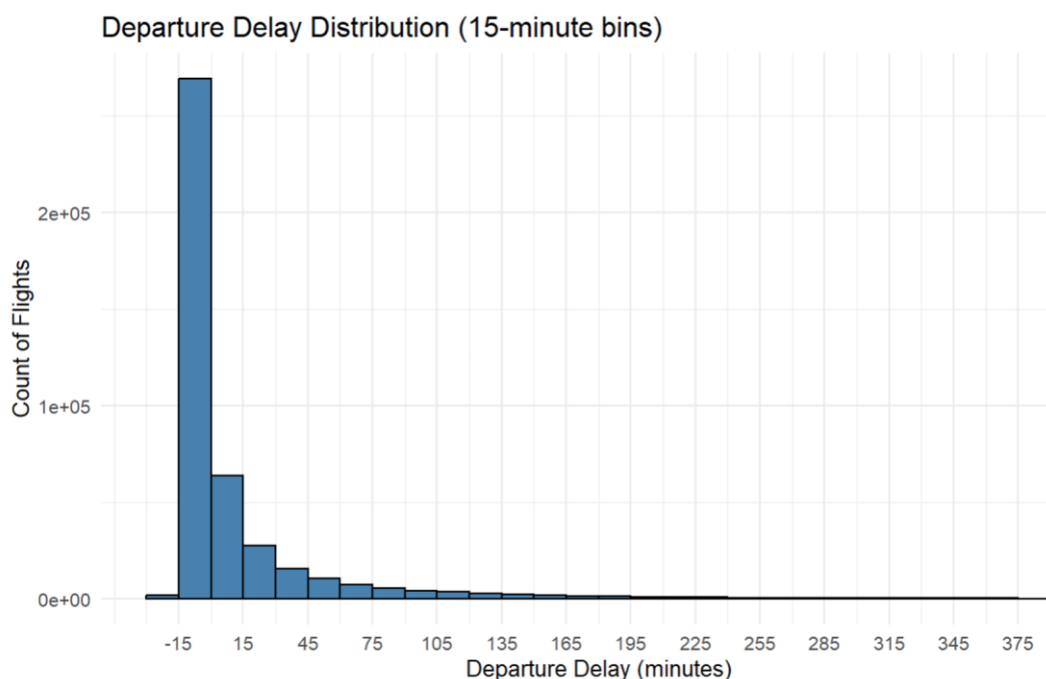
speed. Most gusts fall below 20 mph, while stronger gusts appear relatively rarely. This suggests that extreme gust conditions were uncommon in the dataset.

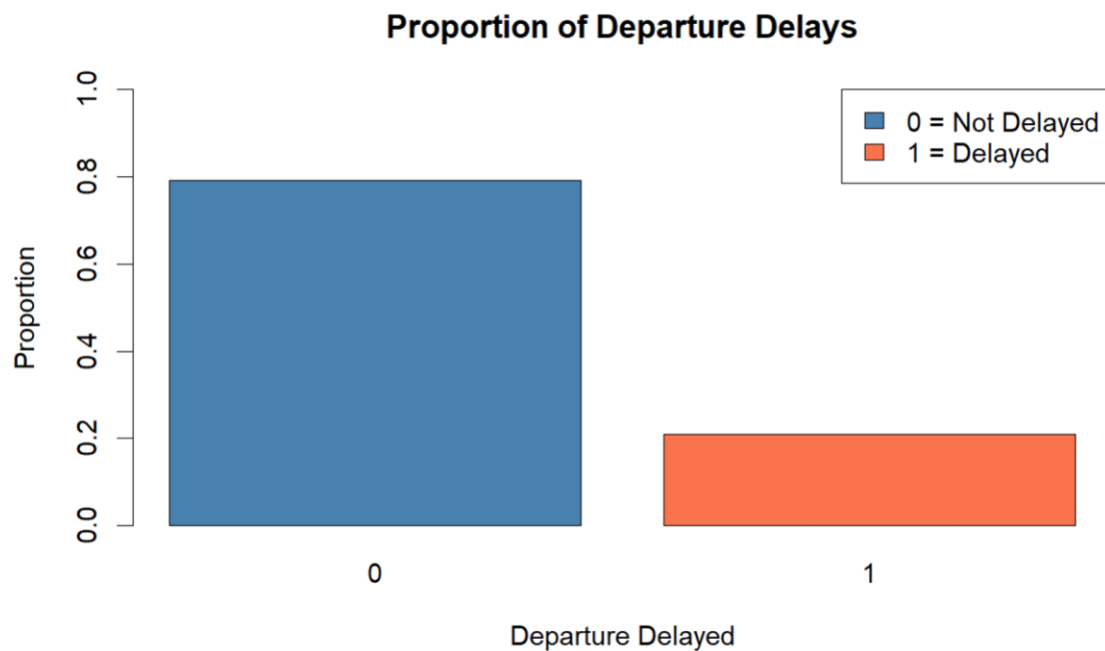
Precipitation: The precipitation distribution is extremely concentrated near zero, with the vast majority of observations showing little or no precipitation. Moderate or heavy precipitation occurs only rarely. This indicates that most flights took place during dry weather conditions.

Pressure: Atmospheric pressure appears to bell-shaped and symmetric, centered around standard pressure values near 1010–1020 millibars. Extremely low-or high-pressure systems are uncommon. This suggests relatively stable atmospheric conditions for most observations.

Visibility: it is heavily concentrated at 10 miles, with relatively few observations at lower visibility levels. This indicates that most flights occurred under clear conditions, while poor visibility events were comparatively rare. Such limited variation may reduce the strength of visibility as a predictor in logistic regression models.

③ Distribution of Response variable





This histogram plots distribution of total departure delays in 15-minute bins. The x-axis represents departure delay in minutes, ranging from about –15 minutes (early departures) to 400 minutes. The y-axis represents the count of flights. Most flights are concentrated near zero delay, with a large spike between –15 and 15 minutes. The distribution is strongly right skewed, with a long tail extending toward large positive delays, indicating that severe departure delays occur less frequently.

The normal vs. delayed departure ratio is about 8:1, which fits the observation that delayed flight combined only occupy a small percent of total flight.

④ About variation of precipitation/visibility

Precipitation <chr>	Frequency <int>	Proportion.Var1 <fctr>	Proportion.Freq <dbl>
0	22747	0	0.8680
1e-04	1715	1e-04	0.0654
0.01	1201	0.01	0.0458
0.02	222	0.02	0.0085
0.03	128	0.03	0.0049
0.04	58	0.04	0.0022
0.05	44	0.05	0.0017
0.06	20	0.06	0.0008
0.07	14	0.07	0.0005
0.08	11	0.08	0.0004

Visibility <chr>	Frequency <int>	Proportion.Var1 <fctr>	Proportion.Freq <dbl>
0.24	6	0.24	0.0002
0.25	56	0.25	0.0021
0.5	54	0.5	0.0021
0.75	42	0.75	0.0016
1	66	1	0.0025
1.25	18	1.25	0.0007
1.5	67	1.5	0.0026
1.75	33	1.75	0.0013
2	214	2	0.0082
2.5	159	2.5	0.0061

Visibility <chr>	Frequency <int>	Proportion.Var1 <fctr>	Proportion.Freq <dbl>
3	365	3	0.0139
3.5	46	3.5	0.0018
4	458	4	0.0175
5	445	5	0.0170
6	431	6	0.0164
7	440	7	0.0168
8	442	8	0.0169
9	575	9	0.0219
10	22267	10	0.8497
NA	23	NA	0.0009

Since the histogram of precipitation and visibility histogram doesn't show clear pattern of variation, and quantitative predictors with very little variation may contribute limited information to a logistic regression model, a closer look into these two variables is needed. Above are tables of distribution of these 2 variables.

The precipitation and visibility distributions are highly unbalanced. Approximately 87% of flights have no precipitation, while only a very small proportion of observations involve moderate or heavy precipitation. Similarly, visibility is heavily concentrated at 10 miles, which accounts for about 85% of the observations, while low-visibility conditions occur relatively rarely. These results suggest that most flights in the dataset occurred under normal weather conditions.

This pattern is important for logistic regression because predictors with very limited variation provide less information for estimating relationships with the response variable. Since precipitation and visibility are concentrated around a few common values, the model may have difficulty detecting strong or stable effects for these variables. As a result, their coefficients may become statistically insignificant or less reliable, not necessarily because weather has no impact on delays, but because severe weather conditions appear infrequently in the dataset.

III: Data Wrangling

Generally, the nycflights23 is a relatively organized dataset with small percent of missing values and required little manipulation for combining different datasets. As explained above, flights is going to be used as the basic dataset.

① Missing Data of All Datasets and Measurements Toward Them

For package nycflights23, missing data is recorded as NA. Here is a brief view of the missing values in all datasets used here.

```
# Number of missing values in each variable
nrow(flights)
colSums(is.na(flights))
```

```
[1] 435352
```

	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	sched_arr_time	arr_delay	carrier
flight	0	0	0	10738	0	10738	11453	0	12534	0
tailnum	0	1913	0	dest	air_time	distance	hour	minute	time_hour	
				0	12534	0	0	0	0	

Both `dep_time` (scheduled departure time in NYC time) and `arr_time`(scheduled arriving time in NYC time) have NA, and that causes the `dep_delay` (scheduled – actual departure time in minutes) and `arr_delay`(scheduled – actual arriving time in minutes) to both missing correspondingly. There are also missing values in `tailnum` and `airtime`.

```
nrow(weather)
colSums(is.na(weather))
...

[1] 26207
   origin      year    month    day      hour    temp    dewp    humid    wind_dir  wind_speed  wind_gust  precip  pressure  visib
   0          0         0      0        0        2        2        2        1225      1041      1041      0         10        23
```

Few values in variables other than wind-related variables are missing. Wind-related variables have similar missing numbers.

	origin	year	month	day	hour	temp	dewp	humid	wind_dir	wind_speed [▲]	wind_gust
17171	LGA	2023	12	18	11	63.0	59.0	86.77	NA	NA	NA
17177	LGA	2023	12	18	17	53.0	48.0	83.06	NA	NA	NA
17178	LGA	2023	12	18	18	50.0	43.0	76.76	NA	NA	NA
17181	LGA	2023	12	18	21	51.0	36.0	56.31	NA	NA	NA
17205	LGA	2023	12	19	21	38.0	21.0	49.98	NA	NA	NA
17208	LGA	2023	12	20	0	35.0	21.0	56.27	NA	NA	NA
17404	LGA	2023	12	28	4	48.0	46.0	92.74	NA	NA	NA
17405	LGA	2023	12	28	5	48.0	46.0	92.74	NA	NA	NA
17407	LGA	2023	12	28	7	48.0	46.0	92.74	NA	NA	NA
17414	LGA	2023	12	28	14	49.0	46.0	89.33	NA	NA	NA
17415	LGA	2023	12	28	15	51.0	48.0	89.42	NA	NA	NA
18904	EWR	2023	3	1	16	44.0	24.1	44.98	NA	NA	NA
18932	EWR	2023	3	2	20	57.0	36.0	45.22	NA	NA	NA
18956	EWR	2023	3	3	20	43.0	27.0	52.91	NA	NA	NA
18978	EWR	2023	3	4	18	49.0	36.0	60.88	NA	NA	NA
18981	EWR	2023	3	4	21	48.0	34.0	58.16	NA	NA	NA
19027	EWR	2023	3	6	19	54.0	17.1	23.08	NA	NA	NA
19029	EWR	2023	3	6	21	52.0	17.1	24.83	NA	NA	NA
19242	EWR	2023	3	15	18	47.0	21.0	35.49	310	NA	NA
19459	EWR	2023	3	24	19	54.0	27.0	35.05	NA	NA	NA
19529	EWR	2023	3	27	17	60.0	23.0	23.81	NA	NA	NA
19530	EWR	2023	3	27	18	56.0	21.0	25.45	NA	NA	NA
	origin	year	month	day	hour	temp	dewp	humid	wind_dir	wind_speed [▲]	wind_gust
8471	JFK	2023	12	19	22	36.0	21.0	54.08	NA	NA	NA
8472	JFK	2023	12	19	23	34.0	20.0	56.12	NA	NA	NA
8473	JFK	2023	12	20	0	33.0	20.0	58.41	NA	NA	NA
8478	JFK	2023	12	20	5	31.0	19.0	60.67	NA	NA	NA
8480	JFK	2023	12	20	7	30.0	18.0	60.53	NA	NA	NA
8514	JFK	2023	12	21	17	41.0	22.0	46.39	NA	NA	NA
8519	JFK	2023	12	21	22	36.0	15.0	41.78	NA	NA	NA
8540	JFK	2023	12	22	19	40.0	11.0	29.94	NA	NA	NA
8678	JFK	2023	12	28	13	52.0	51.0	96.37	NA	NA	NA
8705	JFK	2023	12	29	16	50.0	45.0	82.86	NA	NA	NA
8968	LGA	2023	1	10	15	41.0	25.0	52.61	NA	NA	NA
9024	LGA	2023	1	12	23	53.0	47.0	79.40	NA	NA	NA
9668	LGA	2023	2	8	19	51.0	26.1	37.57	NA	NA	NA
9981	LGA	2023	2	21	20	43.0	35.0	73.45	NA	NA	NA
10135	LGA	2023	2	28	6	33.0	30.0	88.23	NA	NA	NA
10235	LGA	2023	3	4	10	41.0	37.0	85.54	50	NA	NA
10292	LGA	2023	3	6	19	53.0	21.9	29.28	NA	NA	NA
10293	LGA	2023	3	6	20	51.0	18.0	26.69	NA	NA	NA
10299	LGA	2023	3	7	2	47.0	17.1	30.04	NA	NA	NA
10631	LGA	2023	3	20	22	45.0	28.0	51.08	NA	NA	NA
10696	LGA	2023	3	23	15	54.0	44.0	69.09	NA	NA	NA
10845	LGA	2023	3	29	20	54.0	10.9	17.58	NA	NA	NA

These snapshots of a small part of rows with missing data verifies the hypothesis that the wind-related variables tend to have missing values together, and the distribution of time with wind variables missing is not concentrated in fixed months or date or hour. All three airports have missing value in wind-related variables.

```
nrow(planes)
colsums(is.na(planes))
```

```
[1] 4840
      tailnum      year      type manufacturer      model      engines      seats      speed      engine
      0          89          0          0          0          0          0          0          0
```

The only missing value within the planes dataset is the year variable.

	tailnum	year	type	manufacturer	model	engines	seats	speed	engine
4777	N984NK	2023	Fixed wing multi engine	AIRBUS	A320-271N	2	190	0	Turbo-fan
4781	N985NK	2023	Fixed wing multi engine	AIRBUS	A320-271N	2	190	0	Turbo-fan
4790	N987NK	2023	Fixed wing multi engine	AIRBUS	A320-271N	2	190	0	Turbo-fan
6	N102DU	NA	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
10	N103DU	NA	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
16	N104DN	NA	Fixed wing multi engine	AIRBUS	A321-211	2	199	0	Turbo-fan
17	N104DU	NA	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
39	N108DQ	NA	Fixed wing multi engine	C SERIES AIRCRAFT LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
104	N12003	NA	Fixed wing multi engine	BOEING	787-10	2	120	0	Turbo-fan
176	N129DU	NA	Fixed wing multi engine	AIRBUS CANADA LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
234	N139DU	NA	Fixed wing multi engine	AIRBUS CANADA LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
241	N140DU	NA	Fixed wing multi engine	AIRBUS CANADA LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
249	N141DU	NA	Fixed wing multi engine	AIRBUS CANADA LTD PTNRSP	BD-500-1A10	2	133	0	Turbo-fan
406	N17327	NA	Fixed wing multi engine	BOEING	737-8	2	185	0	Turbo-fan
408	N17333	NA	Fixed wing multi engine	BOEING	737-8	2	185	0	Turbo-fan
413	N174US	NA	Fixed wing multi engine	AIRBUS INDUSTRIE	A321-211	2	199	0	Turbo-jet
414	N17529	NA	Fixed wing multi engine	BOEING	737-9	2	48	0	Turbo-fan
415	N17550	NA	Fixed wing multi engine	BOEING	737-9	2	48	0	Turbo-fan
428	N177US	NA	Fixed wing multi engine	AIRBUS INDUSTRIE	A321-211	2	199	0	Turbo-jet
454	N181UW	NA	Fixed wing multi engine	AIRBUS INDUSTRIE	A321-211	2	199	0	Turbo-jet

Considering year only occupies a tiny portion of the entire dataset, and no evidence shows the pattern of missing year value are likely to appear with certain characteristics of other variables (absolute most of aircraft type is fixed wing multi engine), it is reasonable to believe the missing value of this variable appears randomly.

② Data Combination

flights is used as base dataset, weather is combined to flights using identification variables time_hour, month, day, hour, and origin. Those variables share the same name in 2 datasets, so no extra edition is needed; the new dataset is called flights2.

flights2 is then used as base dataset, planes dataset is combined to flights using identification variable tailnum, this variable shares the same name in 2 datasets, so no extra edition is needed, the new dataset is called flights3.

③ Row Dropping Decisions

1. Rows with NA in wind_dir, wind_speed, wind_gust and year are dropped because according to the NA analysis above,

- these variables can be important predictors in the analysis
- the missingness appears approximately random
- and the proportion of missing rows is not too large compared to the overall data number.

2. Rows with dep_time and dep_delay value missing is also dropped, because as explained in NA analysis, whether the flight was delayed is not known and this information cannot be concluded through $\text{scheduled_dep_time} - \text{dep_time} = \text{dep_delay}$. Those rows cannot contribute to the model because the outcome variable is unknown, so removing rows with missing dep_delay is correct and standard practice.

3. Rows with NA in tailnum are dropped because planes dataset relies on tailnum as identification information, having missing value in tailnum will cause variables from planes dataset missing too.

4. As the introduction of planes dataset explains, American Airways (AA) and Envoy Air (MQ) report fleet numbers rather than tail numbers in the talinum column. A fleet number is an internal identifier used by an airline or company to organize and manage aircraft within its own fleet, and they have completely different formats, for example Aircraft 215 instead of normal tail number that is only made up of alphabet and numbers. Information of corresponding sets of fleet number and tail number might be found on these companies' public information page, but it will be difficult to get, create or merge, so when we merge planes to flights, variables inside planes will be missing for these 2 airlines.

Rather than removing these observations, which would exclude a substantial portion of the dataset and introduce selection bias, all flights were retained in the analysis. To address the missing aircraft information, categorical variables (model, type, and manufacturer) were recoded so that missing values were assigned to an "Unknown" category. This allows the model to incorporate aircraft information where available

while still including flights with incomplete data. For the numeric variable year (renamed as plane_year), missing values were imputed using the median aircraft year, which preserves observations while minimizing distortion.

Additionally, to reduce model complexity and avoid overfitting due to a large number of aircraft models, infrequent categories in the model variable were grouped into an "Other Models" category using `fct_lump_min()`. This ensures that the model remains stable and interpretable.

④ Variable Creation

The *dep_delay* variable represents the time between scheduled departure and actual departure time in minutes, so no extra steps are needed to create a quantitative variable for delay. The binary response indicator *dep_delayed* is created using the standard `dep_delay > 15`.

⑤ Now combined Dataset contains 403788 rows and 36 variables, most variables are kept in case of need. No variables are repeated. Check the code appendix for verification.

IV: Generalized Linear Modeling:

Research Question:

What factors best explain if a flight got a departure delay in NYC flights?

Exploratory analysis revealed that both departure and arrival delays are highly right-skewed, with most flights experiencing small delays and a smaller number of extreme delay cases. A strong positive relationship was observed between departure delay and arrival delay, indicating substantial delay propagation throughout the flight process.

Weather variables such as humidity, wind speed, and visibility also showed meaningful associations with delays, with poorer weather conditions generally corresponding to increased arrival delays. Carrier-level differences were evident in both

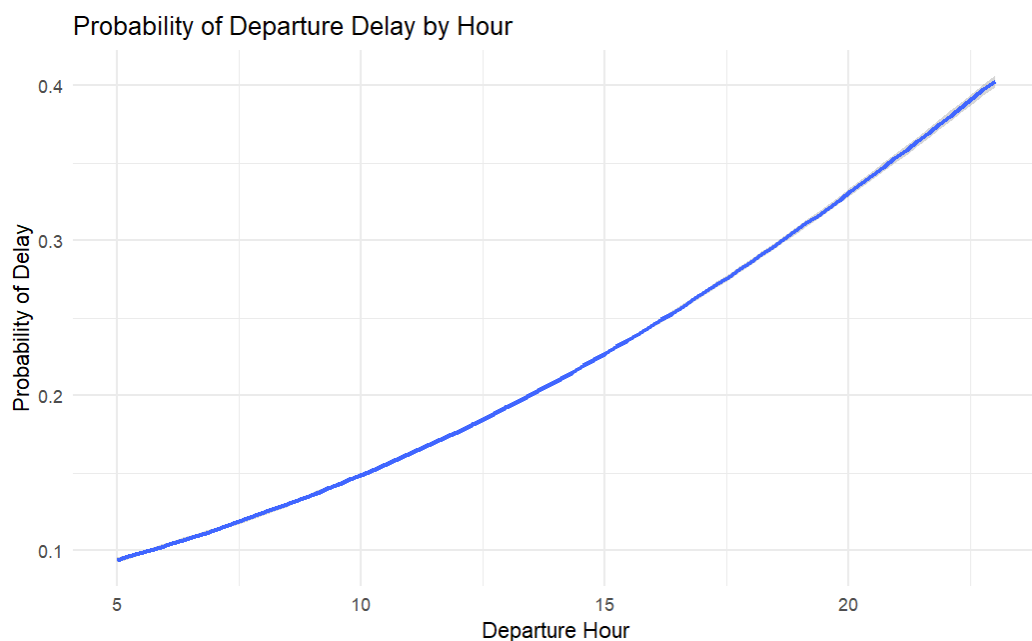
average delay levels and variability, suggesting that airlines differ in operational efficiency. However, most variation in arrival delays occurs at the flight level rather than the airline level, motivating the use of a multilevel modeling framework to appropriately partition variability across levels.

Model 1: Baseline Temporal Model:

The initial model included only departure hour as a predictor:

$$\text{logit}(P(\text{dep_late} = 1)) = \beta_0 + \beta_1(\text{hour})$$

We start with 1 model to compare the improvement other variables made. The coefficient for hour was positive and highly significant ($p < 2e-16$), indicating that flights scheduled later in the day have a higher probability of delay. However, model fit was limited (AIC = 396276), suggesting that additional predictors are necessary.



The curve shows a clear upward trend, meaning congestion and operational load increase delay probability as the day progresses.

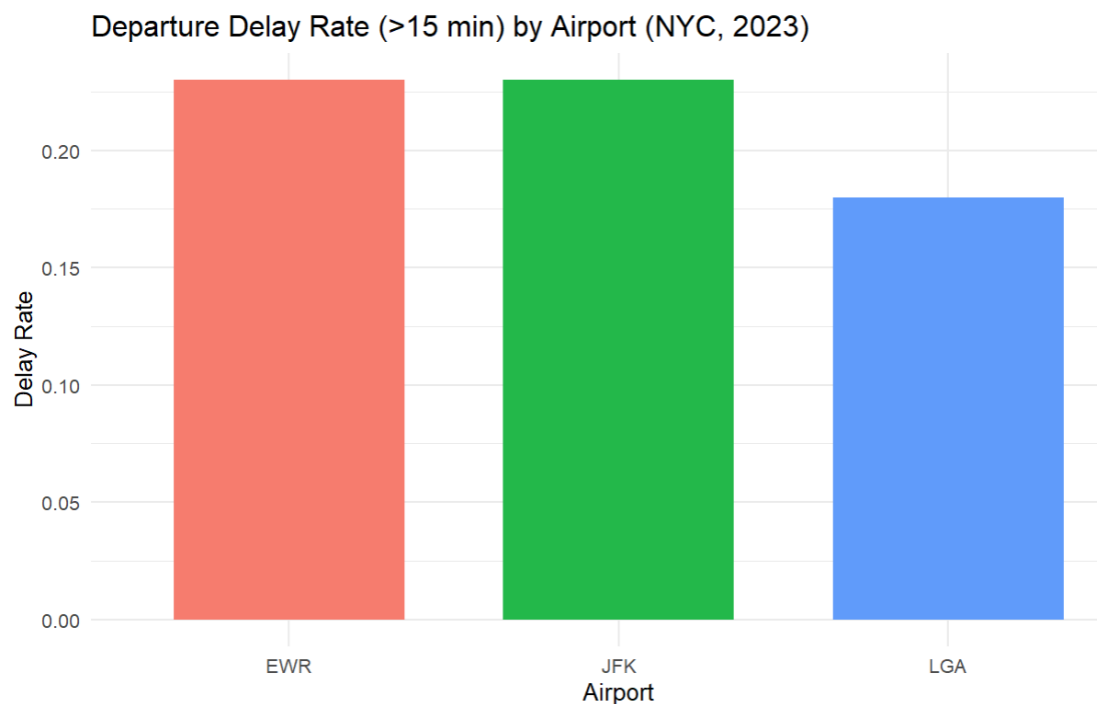
Model 2: Operational + Weather Model

Added operational and weather variables because delays are influenced by airport congestion and environmental conditions.

$$\begin{aligned}\text{logit}(P(\text{dep_late} = 1)) \\ &= \beta_0 + \beta_1(\text{carrier}) + \beta_2(\text{origin}) + \beta_3(\text{hour}) + \beta_4(\text{month}) \\ &\quad + \beta_5(\text{weather variables})\end{aligned}$$

Key results:

- Significant improvements in model fit (AIC = 373981)
- Strong positive effects: humidity, wind speed, precipitation
- Strong negative effect: visibility
- Airline and origin effects are statistically significant
- Seasonal effect (month) is also significant



The results show that EWR has the highest delay rate, with roughly 23% of flights departing more than 15 minutes late. JFK is slightly lower, with a delay rate around 22%, still relatively high. LGA has the lowest delay rate, around 18%, making it the most reliable airport in this comparison.

LaGuardia (LGA) consistently performs better in terms of departure reliability, while

Newark (EWR) is the most delay-prone airport. Even though the difference between EWR and JFK is small, both airports show noticeably higher delay rates than LGA.

To test if the airport is an important factor in influencing delay, a one-way ANOVA is established using confidence level 0.05.

Null hypothesis: The average delay indicator is the same across airports, $\mu_{JFK} = \mu_{LGA} = \mu_{EWR}$

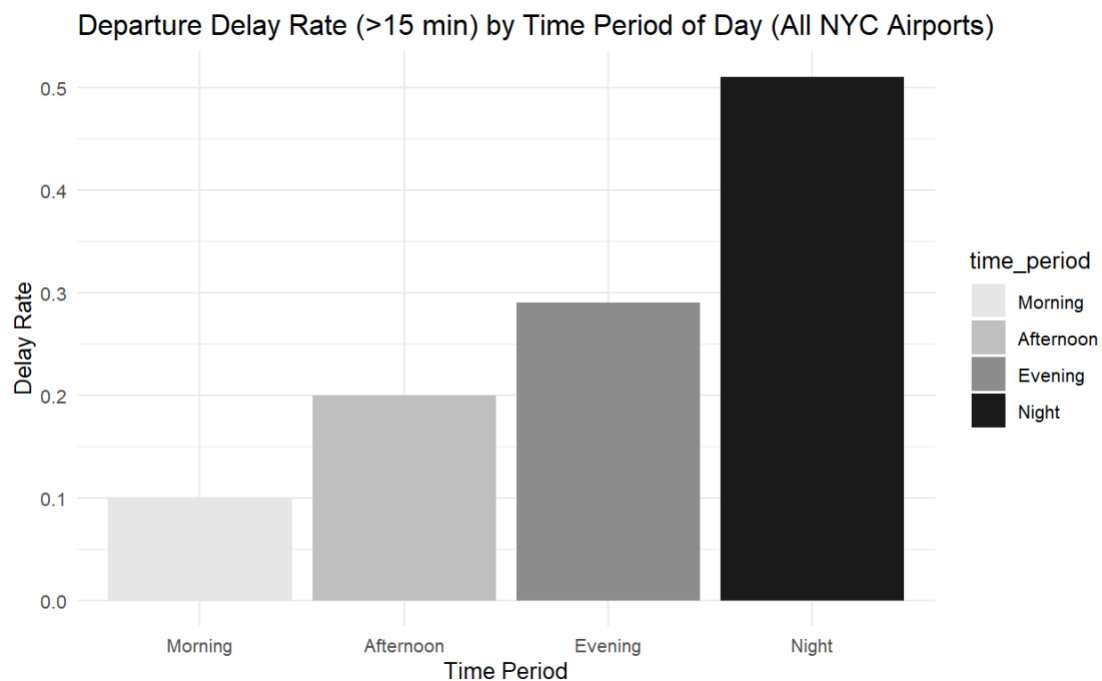
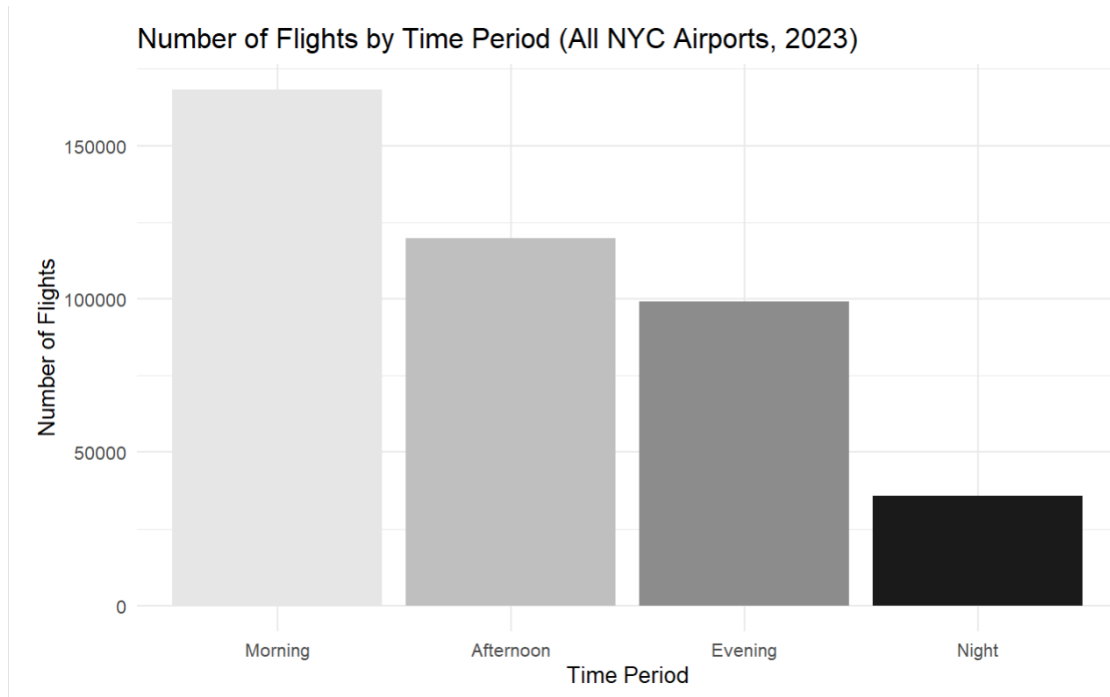
Alternative hypothesis: At least one airport has a different average delay indicator, hereby airport:

```
anova_data <- flights_cat %>%
  mutate(delayed_num = as.numeric(delayed)) %>%
  select(origin, delayed_num)

anova_result <- aov(delayed_num ~ origin, data = anova_data)
summary(anova_result)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
origin          2    196    98.01   593.2 <2e-16 ***
Residuals 424611  70162     0.17
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
10738 observations deleted due to missingness
```

Since the p-value is much smaller than 0.05, the differences we see in delay rates between JFK, LGA, and EWR are too large to reasonably attribute to random chance alone. At confidence level of 95%, we have strong evidence to say airport is an important factor to delay rates for NYC flights.

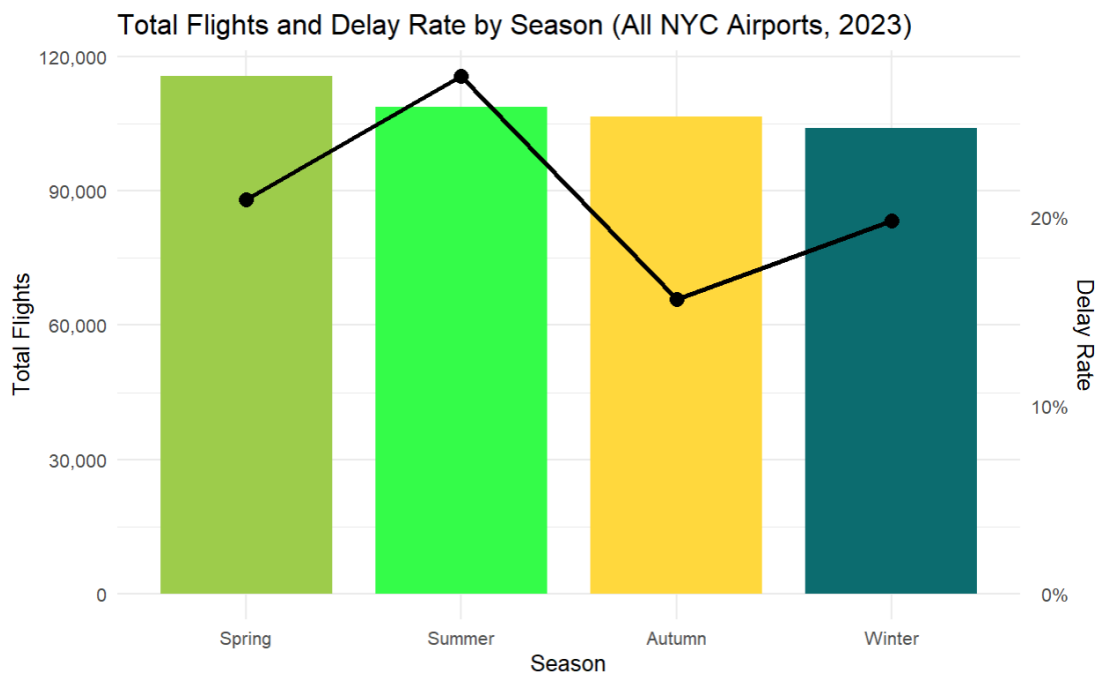


This graph shows how the probability of a departure delay varies across four daily time periods for all NYC airports combined. Flights were grouped into Morning (5:00–11:59), Afternoon (12:00–16:59), Evening (17:00–20:59), and Night (all remaining hours). Note that the total number of hours assigned to each time period is different. However, morning keeps second longest time (7 hours) while holding the lowest delay rate, in comparison, night shares similar length of time with day and is chronologically next to morning, but obviously has higher delay rate; Evening contains only 4 hours but its

delay rate is significantly higher than afternoon, which contains 5 hours. Combined this result with pattern the previous overall flight shows, it is clear that although the night period has the fewest flights, it also showed the highest delay rate in the delay-rate analysis, in contrast, most flights operate in the morning while the delay rate remains lowest. Therefore, this graph does reflect meaningful operational differences rather than unequal time window lengths.

A clear and consistent pattern can be concluded from this graph: departure delays become more likely as the day progresses, with the highest delay risk occurring at night. Specifically, the delay rate was approximately 10% in the morning, increasing to 20% in the afternoon, 29% in the evening, and reaching 51% at night. This means that flights departing at night were about five times more likely to be delayed than flights departing in the morning.

This pattern suggests that delays tend to accumulate throughout the day, By nighttime, the system appears most vulnerable to delays, which strongly increases the chance that a departure will exceed the 15-minute delay threshold.



Here we define season as:

Spring	March to May
Summer	June to August

Fall	September to November
Winter	December to February

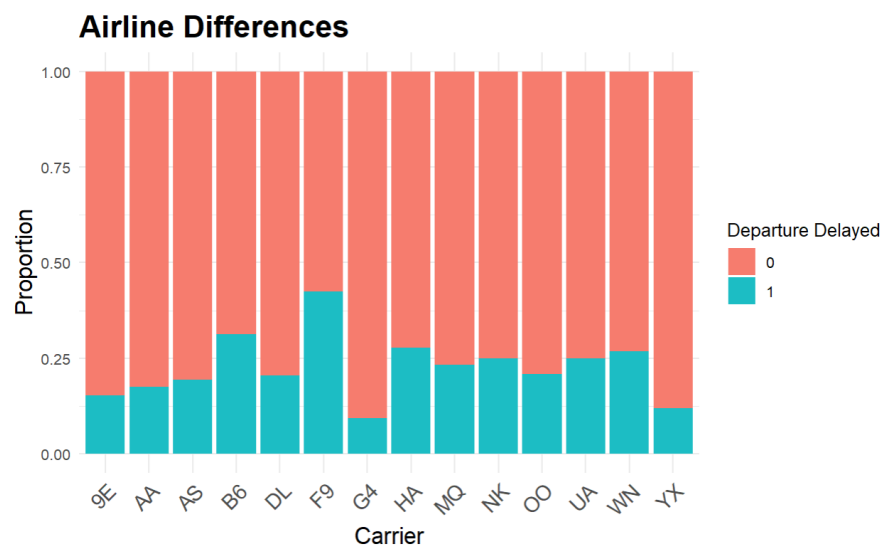
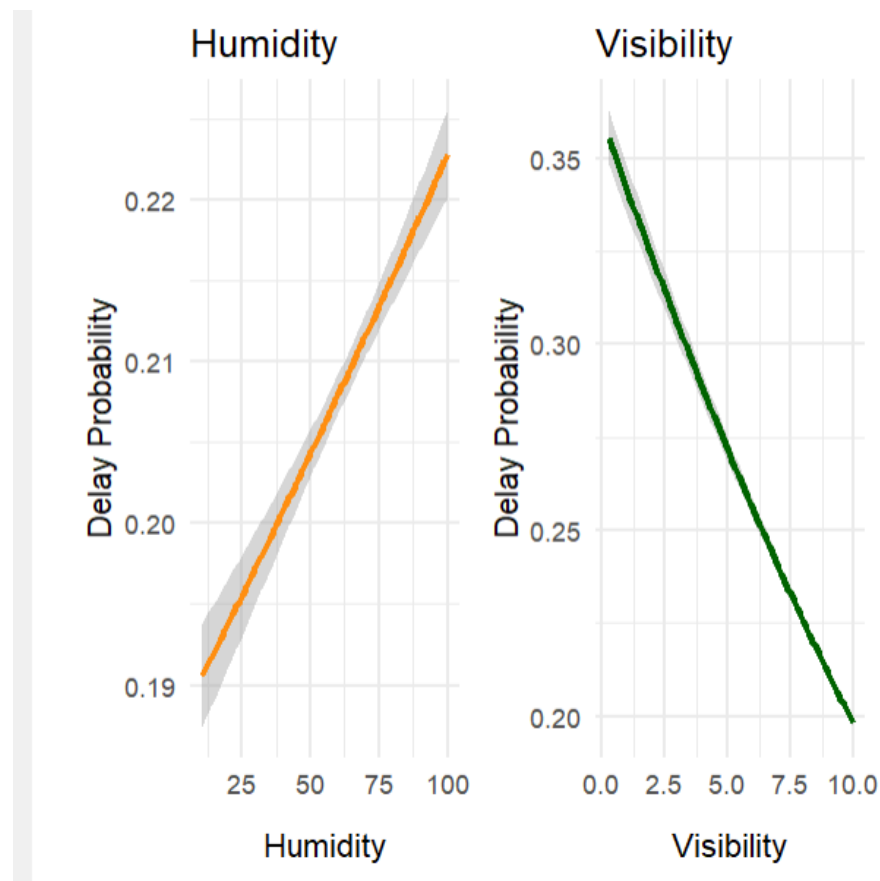
The bar chart shows that the total number of flights across all NYC airports in 2023 is relatively consistent across seasons, with only moderate variation. **Spring has the highest total number of flights**, suggesting that flight schedules and demand increase slightly during the spring travel period. **Summer has the lowest total number of flights**, which may reflect airline scheduling changes, operational constraints, or shifts in demand patterns despite summer being a popular travel season. **Autumn and winter fall in between**, with totals that are very similar to each other. Overall, the differences across seasons are not extreme, meaning that seasonal changes in delay rates (if observed later) are unlikely to be explained simply by one season having dramatically more flights than another.

The line in the graph represents the proportion of flights delayed by more than 15 minutes in each season. The delay rate varies noticeably across the year, with the highest delay rate occurring during the summer. This suggests that flights are more likely to be delayed during the summer months, possibly due to increased travel demand, heavier air traffic, and more frequent weather disruptions such as thunderstorms.

In contrast, the lowest delay rate occurs in autumn, indicating that flights during this season tend to operate more smoothly. Autumn often has milder weather conditions and slightly lower travel volume, which may contribute to fewer delays. Spring and winter show moderate delay rates, falling between the summer peak and the autumn low. Winter delays may be influenced by snow and cold weather, while spring conditions appear relatively stable.

Overall, the graph suggests a seasonal pattern in flight delays, with the highest risk of delay in summer and the lowest risk in autumn, even though the total number of flights does not vary dramatically across seasons.

Therefore, time-related variables such as hour and month should be included.



From these graphs, we can see that:

- Weather conditions such as humidity and visibility have a strong correlation with delay risk
- Airlines differ significantly in delay proportions, so operational structure probably matters beyond time-of-day effects

- Weather conditions and operational variables both can play a major role in explaining departure delays, with precipitation and poor visibility being particularly influential.

So it is reasonable to include operational and temporal variables in the model.

Model 3: Full Extended Model (Including Aircraft Type)

Added aircraft model variables to test whether aircraft structure contributes additional explanatory power:

$$\text{logit}(P(\text{dep_late} = 1)) = \beta_0 + \text{carrier} + \text{origin} + \text{time variables} + \text{weather variables} + \text{aircraft model}$$

Key results:

- Best model fit among all models (AIC = 373086)
- Several aircraft models are significant predictors of delay probability
- Operational and weather effects remain stable and significant
- Some aircraft types (e.g., certain Boeing and Airbus models) show higher delay risk

Performance of models: Which Model is better?

That depends on what kind of model the user wants. Statistically, model3 has better AIC and ROC performance.

	df <dbl>	AIC <dbl>
model1	2	396276.7
model2	23	373981.1
model3	90	373086.7

```

prob1 <- predict(model1, type = "response")
prob2 <- predict(model2, type = "response")
prob3 <- predict(model3, type = "response")

roc1 <- roc(flights3$dep_delayed, prob1)
roc2 <- roc(flights3$dep_delayed, prob2)
roc3 <- roc(flights3$dep_delayed, prob3)

auc(roc1)
auc(roc2)
auc(roc3)

plot(roc1,
     col = "blue",
     main = "ROC Curves for Logistic Regression Models")

plot(roc2,
     col = "red",
     add = TRUE)

plot(roc3,
     col = "darkgreen",
     add = TRUE)

legend("bottomright",
      legend = c(
        paste("Model 1 AUC =", round(auc(roc1), 3)),
        paste("Model 2 AUC =", round(auc(roc2), 3)),
        paste("Model 3 AUC =", round(auc(roc3), 3))
      ),
      col = c("blue", "red", "darkgreen"),
      lwd = 2)

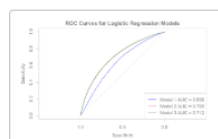
```

```

setting "levels": control = 0, case = 1
setting direction: controls < cases
setting "levels": control = 0, case = 1
setting direction: controls < cases
setting "levels": control = 0, case = 1
setting direction: controls < cases
Area under the curve: 0.6377
Area under the curve: 0.7093
Area under the curve: 0.7118

```

R Console

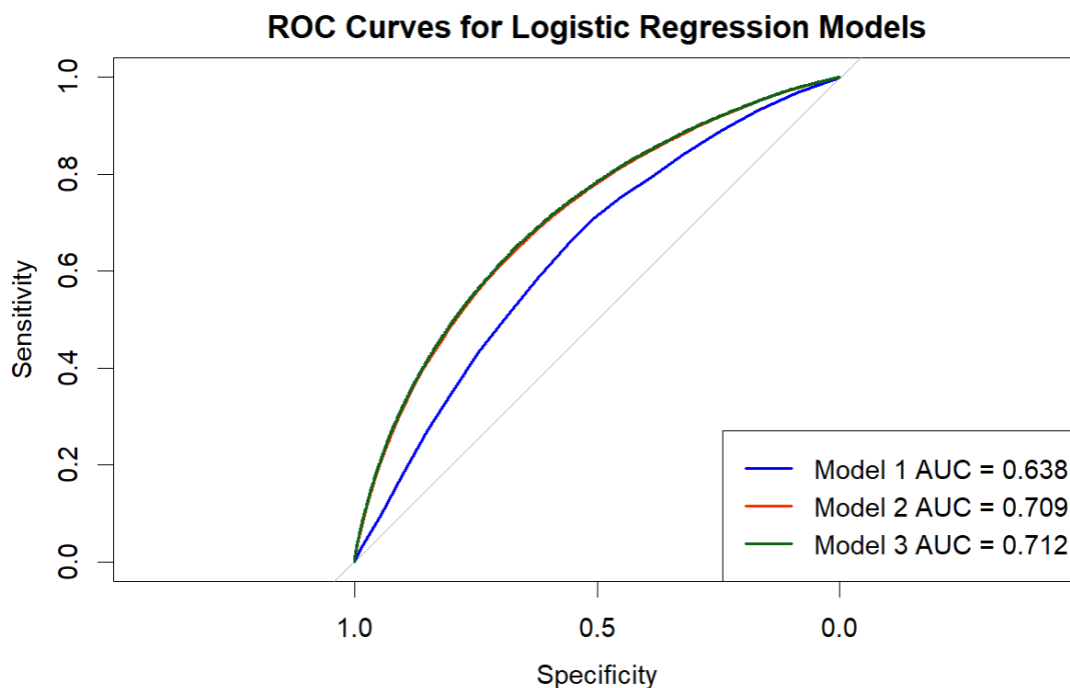
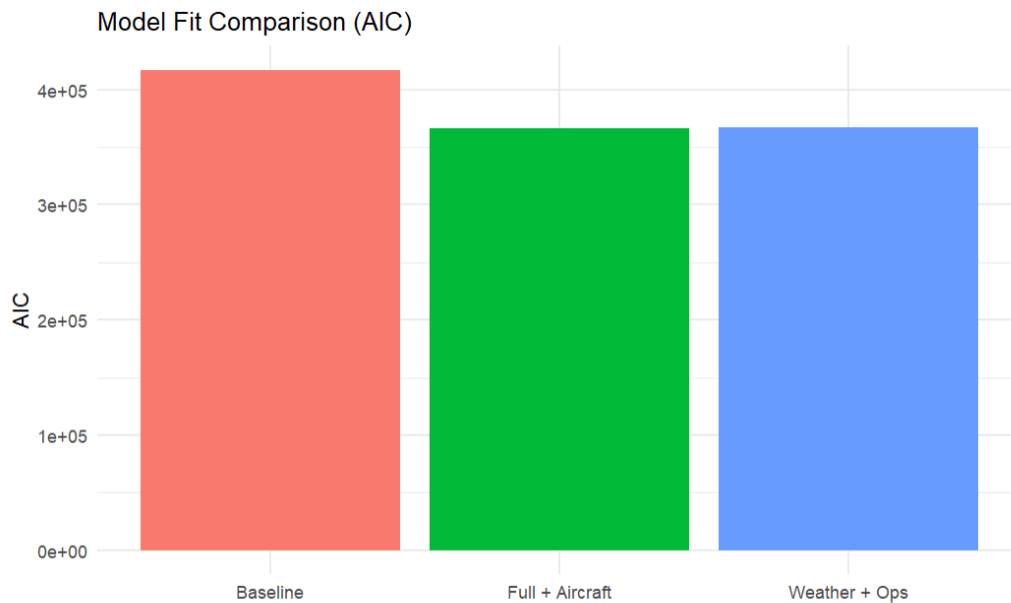


```

setting levels: control = 0, case = 1
setting direction: controls < cases
setting levels: control = 0, case = 1
setting direction: controls < cases
setting levels: control = 0, case = 1
setting direction: controls < cases

```

Area under the curve: 0.6377
Area under the curve: 0.7093
Area under the curve: 0.7118



However, the improvement from Model 2 to Model 3 is relatively modest compared to the large increase in complexity. Model 3 adds about 67 additional aircraft model categories, which makes the model more difficult to interpret and communicate. Despite this, the added complexity is not as problematic as adding 67 unrelated variables because each flight can only belong to one aircraft model category at a time. In practice, this means the model is still computationally manageable, and the aircraft model acts as a single categorical predictor rather than dozens of independent

measurements.

Model Comparison Summary

Model	Predictors Included	AIC	ROC
Model 1	Hour only	416,908	0.6377
Model 2	Carrier + origin + weather + time	374,845	0.7093
Model 3	Model 2 variables + Aircraft model	366,462	0.7188

For practical interpretation, Model 2 may be preferable because it achieves nearly the same predictive performance while remaining much simpler. The variables in Model 2, such as carrier, airport, time, humidity, precipitation, and visibility, are easier for passengers and airport operators to understand directly. Therefore, Model 2 provides a strong balance between prediction accuracy and interpretability, while Model 3 is most useful when maximum predictive performance is the primary goal.

modelA320-271N	0.3888799	0.0811493	4.792	1.65e-06	***
modelA321-211	0.5727580	0.0566923	10.103	< 2e-16	***
modelA321-231	0.1724762	0.0642933	2.683	0.007304	**
modelA321-253N	0.8345693	0.1145561	7.285	3.21e-13	***
modelA321-253NX	0.4499965	0.1000018	4.500	6.80e-06	***
modelA321-271NX	0.4992797	0.0646827	7.719	1.17e-14	***
modelA330-243	-0.9261823	0.5632856	-1.644	0.100125	
modelA330-941	0.0759970	0.2111721	0.360	0.718936	
modelBD-500-1A10	0.4830222	0.0596589	8.096	5.66e-16	***
modelCL-600-2D24	0.6597476	0.2524334	2.614	0.008961	**
modelERJ 170-100 LR	0.4931919	0.1037562	4.753	2.00e-06	***
modelERJ 170-100 SE	0.5019099	0.0878231	5.715	1.10e-08	***
modelERJ 170-100SU	0.5089426	0.0988946	5.146	2.66e-07	***
modelERJ 170-200 LL	0.7970122	0.1770495	4.502	6.74e-06	***
modelERJ 170-200 LR	0.4172155	0.0833262	5.007	5.53e-07	***
modelERJ 190-100 IGW	0.1570436	0.0650844	2.413	0.015825	*
modelUnknown	0.3678199	0.0594326	6.189	6.06e-10	***
modelOther Models	0.6656122	0.1046162	6.362	1.99e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

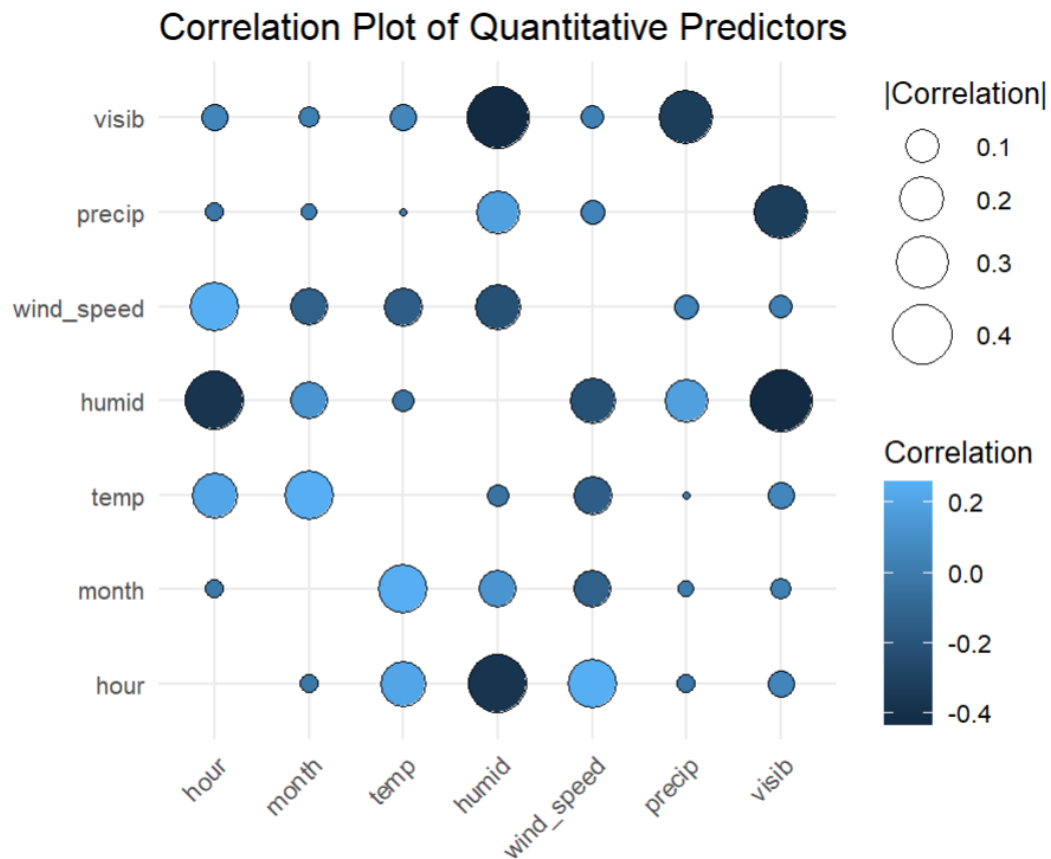
Many intercepts of aircraft models also show p value much lower than 0.05, which also shows the necessity of adding the model variable.

In conclusion, including aircraft type improves model fit, as shown by the lower AIC value, Predictive performance also improves consistently as additional operational and environmental variables are introduced. The results highlight that departure delays are not driven by a single factor, but rather by a combination of:

- Time-of-day congestion effects
- Airline and airport operational differences
- Weather conditions
- Aircraft-specific characteristics

The final model demonstrates the strongest explanatory power, indicating that delay risk is a multi-layered phenomenon influenced by both environmental and structural aviation factors, but in case of seeking convenience, model 2 is also recommended.

Risk of Multicollinearity?



The correlation plot shows that several quantitative predictors have moderate relationships with each other, particularly humidity with visibility, humidity with precipitation, and temperature with hour. However, most correlations appear to remain in the weak-to-moderate range rather than extremely high levels. This suggests that some multicollinearity may exist among the weather variables, which is expected because weather conditions are naturally related, but the correlations are likely not severe enough to create major instability in the logistic regression model. Overall, the plot indicates that the predictors still contain distinct information and can reasonably be included together in the analysis.

```
vif(model2)
vif(model3)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
carrier	2.519319	13	1.036177
origin	2.702198	2	1.282122
hour	1.344825	1	1.159666
month	1.199325	1	1.095137
temp	1.254512	1	1.120050
humid	1.749934	1	1.322851
wind_speed	1.209664	1	1.099847
precip	1.129151	1	1.062615
visib	1.431984	1	1.196655
	GVIF	Df	GVIF ^{1/(2*Df)}
carrier	1.043008e+09	13	2.222579
origin	3.787666e+00	2	1.395060
hour	1.359810e+00	1	1.166109
month	1.214280e+00	1	1.101944
temp	1.264509e+00	1	1.124504
humid	1.750921e+00	1	1.323224
wind_speed	1.210047e+00	1	1.100022
precip	1.129369e+00	1	1.062718
visib	1.432309e+00	1	1.196791
model	1.011891e+09	67	1.167354

The VIF results also supported that multicollinearity is not a major issue in either logistic regression model. In both models, all adjusted GVIF values are relatively small, with the largest being about 2.22 for carrier in Model 3. Since these values are well below common concern thresholds such as 5 or 10, the predictors do not appear to have problematic linear dependence with one another.

This indicates that the coefficient estimates in the models are likely stable and interpretable, and that adding variables such as model did not introduce severe multicollinearity. Therefore, the regression results are less likely to suffer from inflated standard errors caused by strong relationships among predictors.

V: Multilevel Modeling

Research question: How do flight-level factors affect arrival delays, and how do these relationships vary across airlines?

The data have a hierarchical structure:

- **Level 1 (Flight level):** individual flight characteristics (e.g., departure delay, arrival delay, distance, time of day, weather conditions)
- **Level 2 (Carrier level):** airline carriers (14 unique airlines), which group flights and capture the airline variation

This nested structure justifies this dataset can be used to construct a multilevel model.

Although no separate airline-level covariates are available, multilevel modeling is still appropriate because flights are naturally grouped within airlines. The random effect's structure allows the model to account for between-airline variability in arrival delays.

Explanation of variable selection:

In the MLM, `arr_delay` was used as the response variable, and predictors included operational and weather-related variables such as departure delay, departure hour, month, origin of airport, humidity, visibility, wind speed, and precipitation. Because flights are naturally grouped within airlines, we treated carrier as the Level 2 grouping variable in the model.

In addition to the fixed effects, we included random effects for airlines in order to allow the baseline arrival delay relationship to vary across carriers. This approach helps account for the possibility that some airlines consistently experience larger or smaller delays than others, even after controlling weather and scheduling conditions. After fitting the model, we examined the fixed-effect estimates, random-effect variability, and overall model fit statistics to determine which variables were most strongly associated

with arrival delay and whether airline-level variation meaningfully contributed to the models.

Established Model:

```

Formula: arr_delay ~ dep_delay + hour + month + origin + temp + humid + wind_speed + precip + visib + (1 | carrier)
Data: flights3

REML criterion at convergence: 3467450

Scaled residuals:
    Min       1Q   Median       3Q      Max
-5.2494 -0.5949 -0.1004  0.4594 17.9587

Random effects:
Groups Name Variance Std.Dev.
carrier (Intercept) 7.653 2.766
Residual 326.578 18.071
Number of obs: 401936, groups: carrier, 14

Fixed effects:
              Estimate Std. Error t value
(Intercept) -1.355e+01 8.039e-01 -16.860
dep_delay    1.009e+00 5.426e-04 1859.473
hour         5.940e-02 7.089e-03 8.379
month       -4.148e-01 8.741e-03 -47.451
originJFK   -3.369e+00 9.881e-02 -34.098
originLGA   -2.314e+00 8.793e-02 -26.321
temp         6.306e-02 2.010e-03 31.373
humid        9.732e-02 1.873e-03 51.951
wind_speed   1.723e-01 5.927e-03 29.076
precip       2.281e+01 3.437e+00 6.637
visib       -3.263e-01 1.748e-02 -18.669

Correlation of Fixed Effects:
(Intr) dp_dly hour month orgJFK orgLGA temp humid wnd_sp precip
dep_delay 0.022
hour      -0.148 -0.127
month     -0.002 0.066 -0.016
originJFK -0.003 0.022 -0.076 0.003
originLGA -0.061 0.015 0.023 -0.017 0.657
temp      -0.093 -0.090 -0.242 -0.242 0.039 0.006
humid     -0.269 -0.078 0.377 -0.146 -0.201 0.012 -0.025
wind_speed -0.092 -0.030 -0.201 0.055 -0.143 -0.072 0.173 0.165
precip    -0.037 -0.013 -0.002 -0.011 0.012 -0.002 -0.024 -0.071 -0.078
visib     -0.267 0.014 0.141 -0.082 -0.093 -0.010 -0.062 0.435 0.025 0.258

fit warnings:
Some predictor variables are on very different scales: consider rescaling.
You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.

```

We got the equation as:

$$\text{arr_delay}_{ij} = \beta_0 + \beta_1 (\text{dep_delay}_{i,j}) + \beta_2 (\text{hour}_{i,j}) + \beta_3 (\text{month}_{i,j}) + \beta_4 (\text{origin}_{i,j}) + \beta_5 (\text{temp}_{i,j}) + \beta_6 (\text{humid}_{i,j}) + \beta_7 (\text{wind_speed}_{i,j}) + \beta_8 (\text{precip}_{i,j}) + \beta_9 (\text{visib}_{i,j}) + u_{0j} + \varepsilon_{ij}$$

where:

- i = individual flight
- j = airline carrier
- u_{0j} = random intercept for airline j , representing airline-specific differences in average arrival delay
- ε_{ij} = remaining flight-level error term

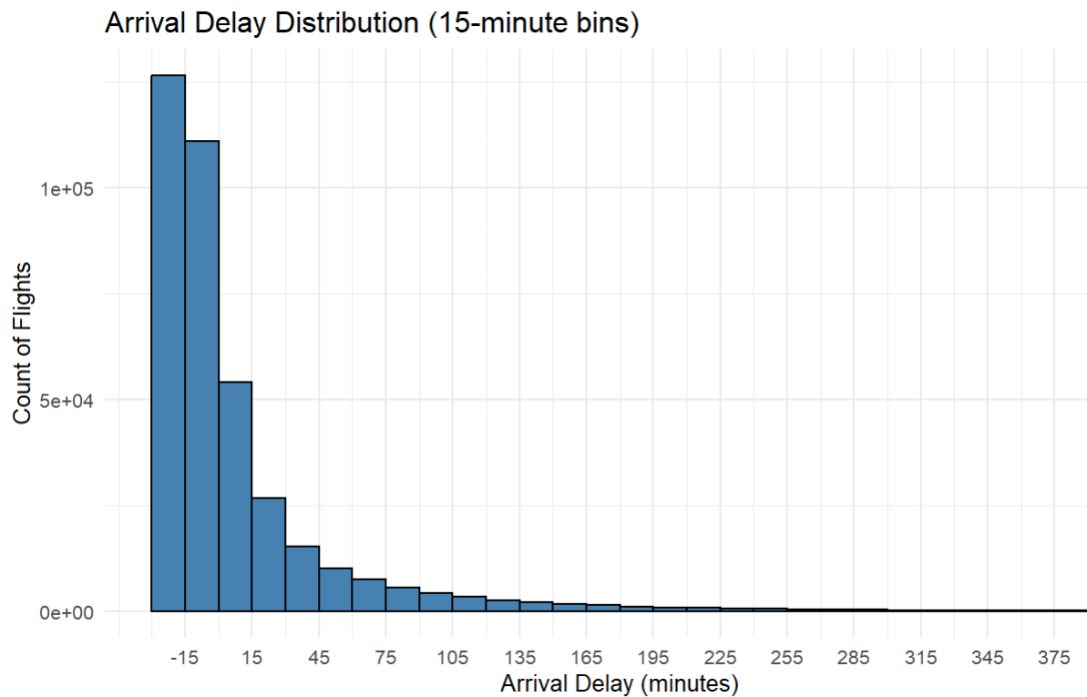
Output Interpretation:

The random effects output shows that the variance of the random intercept for carrier is **7.653**, with a standard deviation of **2.766**. This means that airlines differ in their baseline arrival delay levels, even after accounting for departure delay, time, origin, and weather variables. In other words, some airlines tend to have slightly higher or lower average arrival delays than others. Because there is measurable variation across carriers, using a multilevel model with airline as the Level 2 grouping variable is appropriate. However, the residual variance (**326.578**) is much larger than the carrier-level variance, so most of the variation in arrival delay still occurs at the individual flight level.

The fixed effects results show that departure delay is the strongest predictor of arrival delay, with a large positive coefficient and extremely large t-value. This indicates that flights departing late are much more likely to arrive late as well. Weather variables such as humidity, wind speed, and precipitation also have positive relationships with arrival delay, while visibility has a negative relationship, suggesting that poor weather conditions increase arrival delays. In addition, the significant origin of airport effects indicate that arrival delay patterns differ across NYC airports even after controlling weather and scheduling variables.

Overall, the multilevel model suggests that both flight-level operational factors and airline-level grouping structure contribute to arrival delays. While most variation occurs at the flight level, the random carrier effect shows that airlines still differ in their baseline delay behavior. This supports the use of multilevel modeling for understanding how delay relationships vary across airlines.

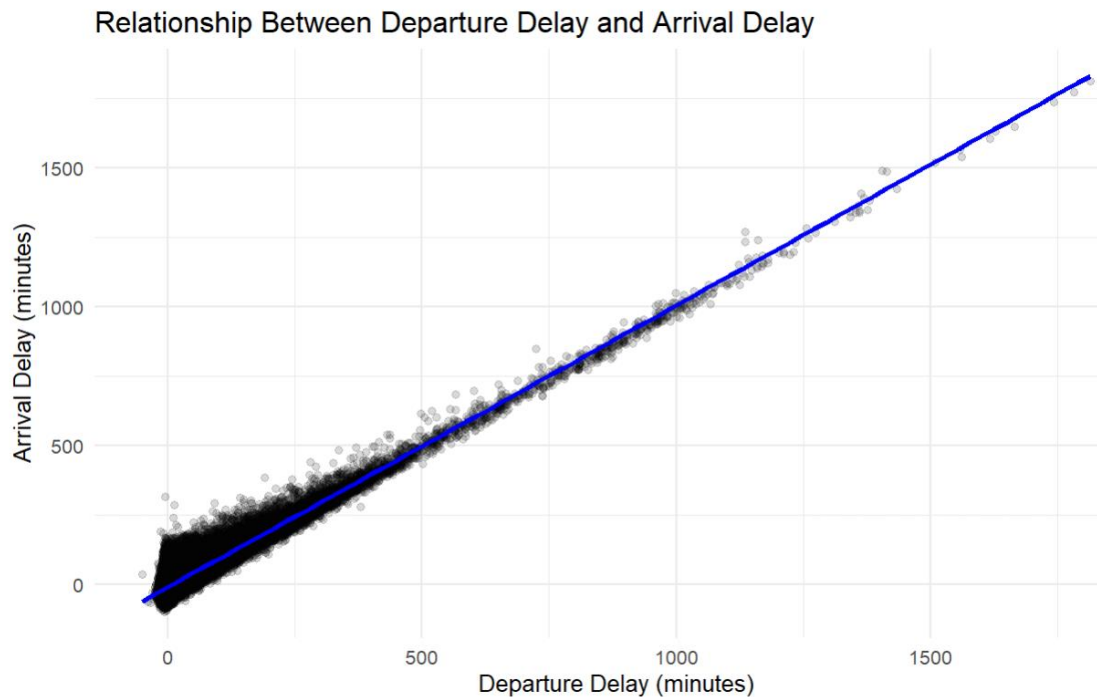
Arrival Delay Patterns:



The distribution of arrival delays is strongly right skewed. Most flights arrive close to schedule or with only small delays, while a smaller number of flights experience very large delays exceeding 100 minutes. This indicates that arrival delay behavior is not evenly distributed across flights and suggests that multiple operational and environmental factors may contribute to extreme delays. Because flights are grouped within airlines, and airlines may have different scheduling practices and operational efficiency, a multilevel model is appropriate for capturing both flight-level effects and airline-level variation in arrival delays.

Extra Exploration: Is there a strong correlation between the rate of departure and arrival delay time?

The total departure and arrival delay rate of flights departed from NYC airports shares a very similar value, which is around 20.8%. One concern raised based on this observation: is departure delay time a significant indicator of arrival delay time? A scatter plot is made to answer this concern preliminarily.



Each point in this scatter plot represents one flight, and the fitted regression line represents the simple linear regression model predicting arrival delay from departure delay.

The plot reveals a strong positive linear relationship between departure delay and arrival delay. As departure delay increases, arrival delay also increases, and the points closely follow the fitted line. This indicates that departure delay is a strong predictor of arrival delays. In other words, flights that leave late are very likely to arrive late as well. The slope of the fitted line appears close to 1, suggesting that for each additional minute of departure delay, the arrival delay increases by approximately one minute on average. This means that most flights do not fully recover lost time during the trip, although some flights may reduce the delay slightly.

Overall, the scatterplot and fitted regression line indicate that departure delay and arrival delay have a strong linear association, making a simple linear regression model appropriate for describing their relationship. Therefore, a t-test and overall-F test are run to check variable importance, taking confidence level 0.95.

```

call:
lm(formula = arr_delay ~ dep_delay, data = delay_scatter)

Residuals:
    Min       1Q   Median       3Q      Max
-97.08 -10.70  -1.65    8.56  330.62

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.557465    0.029118  -328.2   <2e-16 ***
dep_delay    1.014823    0.000522  1944.3   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.35 on 422816 degrees of freedom
Multiple R-squared:  0.8994,    Adjusted R-squared:  0.8994
F-statistic: 3.78e+06 on 1 and 422816 DF,  p-value: < 2.2e-16

```

The p-value for the slope is less than 2×10^{-16} , which is far below 0.05 threshold. Therefore, we reject the null hypothesis and conclude that departure delay is a statistically significant predictor of arrival delay. R^2 gives 0.8994, which is an incredibly high value for a model made up of one variable alone, implying about 90% of variance of arrival delay time is explained by departure delay time.

The slope is about 1.015, meaning for each additional 1 minute of departure delay, the arrival delay increases by about 1.01 minutes on average.

This suggests that flights usually do not fully make up lost time once they depart late. Travelers should expect flights that departed late to arrive late by a very similar amount of time.

In conclusion, the multilevel model shows that flight-level operational and weather variables strongly affect arrival delays, while airlines also differ in their baseline delay behavior. Departure delay was the strongest predictor, and weather conditions such as humidity, wind speed, and visibility also contributed significantly. The random intercept results indicate that accounting for airline-level grouping improves the modeling of arrival delays.

Performance Comparison between multilevel model and standard linear regression model

```
lm1 <- lm(
  arr_delay ~ dep_delay + hour + month + origin +
  temp + humid + wind_speed + precip + visib,
  data = flights3
)

AIC(lm1, m1m1)
```

Description: df [2 × 2]

	df <dbl>	AIC <dbl>
lm1	12	3470658
m1m1	13	3467476

The multilevel model (m1m1) has a slightly smaller AIC value than the standard linear regression model (lm1) with AIC. Since lower AIC values indicate better model fit, this suggests that including airline carriers as a random effect improves the model. Although the improvement is not extremely large, the results indicate that there is meaningful airline-level variation in arrival delays that is not fully explained by the flight-level predictors alone. Therefore, the multilevel model is preferred because it better accounts for the grouped structure of flights within airlines than standard linear regression.

VI: Machine Learning:

After consideration, Random Forest regression is chosen to further investigate our research question, because the research question focuses on:

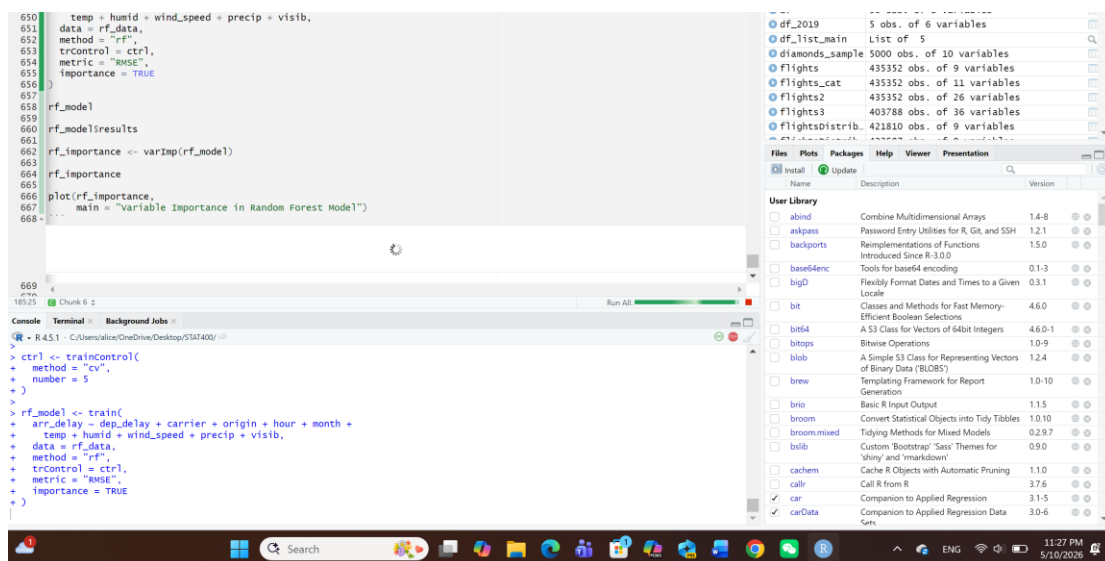
- understanding predictors,
- comparing importance of variables,
- and improving prediction of arrival delays.

While Random Forest does a good job on:

- handling nonlinear relationships,
- capturing interactions automatically,
- working with many predictors,
- and provides variable importance rankings.

Therefore, I believe it will complement the multilevel model nicely.

That was the plan, however, somehow my R studio could not complete executing my RF coding and gave an outcome, so I was forced to change a method.



Complete code:

```

rf_data <- flights3 %>%
  select(
    arr_delay,
    dep_delay,
    carrier,
    origin,
    hour,
    month,
    temp,
    humid,
    wind_speed,
    precip,
    visib
  ) %>%
  na.omit()

set.seed(123)

ctrl <- trainControl(
  method = "cv",
  number = 5
)

rf_model <- train(
  arr_delay ~ dep_delay + carrier + origin + hour + month +
  temp + humid + wind_speed + precip + visib,
  data = rf_data,
  method = "rf",
  trControl = ctrl,
  metric = "RMSE",
  importance = TRUE
)

rf_model

rf_model$results

rf_importance <- varImp(rf_model)

rf_importance

plot(rf_importance,
     main = "Variable Importance in Random Forest Model")

```

Then I tried LASSO Regression because:

- it handles many predictors,
- reduces overfitting.

```
ml_vars <- c(
  "arr_delay",
  "dep_delay",
  "carrier",
  "origin",
  "hour",
  "month",
  "temp",
  "humid",
  "wind_speed",
  "precip",
  "visib"
)

ml_data <- flights3[, ml_vars]

ml_data <- na.omit(ml_data)

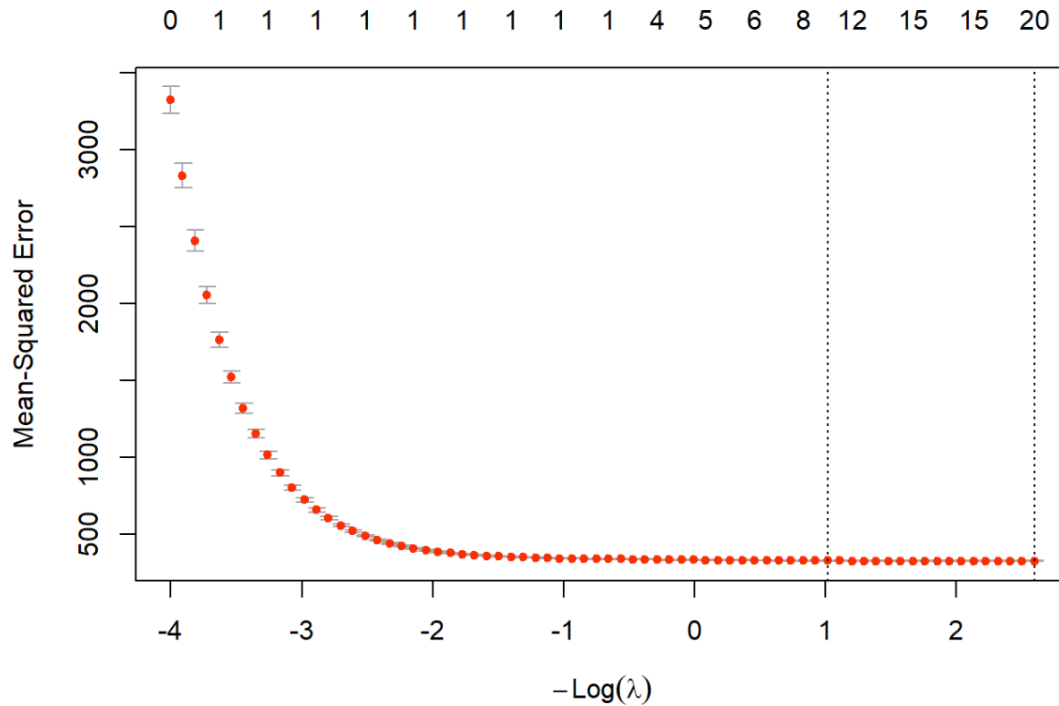
x <- model.matrix(
  arr_delay ~ dep_delay + carrier + origin +
    hour + month + temp + humid +
    wind_speed + precip + visib,
  data = ml_data
)[, -1]

y <- ml_data$arr_delay

set.seed(123)

lasso_cv <- cv.glmnet(
  x = x,
  y = y,
  alpha = 1,
  nfolds = 5
)

plot(lasso_cv)
```



The cross-validation plot shows that prediction error decreases rapidly as more useful predictors are added into the LASSO model. At large values of λ , the penalty is strong, and only a few variables remain in the model, resulting in relatively high mean squared error. As the penalty becomes smaller, additional predictors are included, and the prediction error drops substantially before eventually leveling off. This pattern suggests that several flight-level and weather-related variables provide important information for explaining arrival delays. However, after a certain point, adding more variables produces only very small improvements in prediction of accuracy.

The dashed vertical lines represent the two commonly used choices of $\lambda_{\min}$, which gives the lowest prediction error, which selects a slightly simpler model with nearly the same performance. The relatively flat shape of the curve near the minimum indicates that simpler models may perform almost as well as more complicated ones. This supports the idea that a smaller set of key predictors can still explain most variation in arrival delays.

```
lasso_cv$lambda.min
```

```
## [1] 0.07415858
```

```
coef(lasso_cv, s = "lambda.min")
```

```
## 24 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept) -12.91449944
## dep_delay   1.00830329
## carrierAA    0.11913807
## carrierAS   -2.49351710
## carrierB6    0.81763446
## carrierDL   -3.85455462
## carrierF9    .
## carrierG4   -0.35053406
## carrierHA    4.91110811
## carrierMQ    .
## carrierNK    .
## carrierOO    2.69668655
## carrierUA   -0.13138853
## carrierWN   -0.15510775
## carrierYX    0.09174199
## originJFK   -2.39815195
## originLGA   -1.60150544
## hour        0.03979926
## month       -0.38647886
## temp        0.05726436
## humid       0.08865557
## wind_speed  0.15068717
## precip     17.89772244
## visib      -0.33025240
```

This output shows the coefficients selected by the LASSO model at the optimal penalty value, $\lambda_{\min} = 0.0742$. LASSO shrinks less important coefficients toward zero, so variables with coefficients equal to "." were effectively removed from the model. In this case, variables such as carrierF9, carrierMQ, and carrierNK were not retained, suggesting they did not contribute much additional predictive value after accounting for the other variables.

The results also suggest that departure delay is by far the strongest predictor of arrival delay, with a coefficient near 1.01. This indicates that larger departure delays are strongly associated with larger arrival delays. Weather variables also remain important in the model. For example, precipitation has a large positive coefficient, while visibility has a negative coefficient, meaning worse weather conditions tend to increase arrival delays. Humidity and wind speed also show positive relationships with delay.

Several airline indicators remain in the model as well, showing that differences across airlines still matter even after controlling weather and operational variables. Some carriers have positive coefficients while others have negative coefficients, suggesting that airlines differ in average delay performance. Airport origin variables (originJFK and originLGA) were

also retained, indicating that departure airport contributes to explaining arrival delay patterns.

Overall, the LASSO model's result supports the earlier regression and multilevel modeling results: arrival delays are influenced by a combination of operational factors, weather conditions, airport effects, and airline-specific differences.

VII: Conclusion and Concerns

After building various models and graphs to explore and answer questions related to delay using data from 2023, now we are able to address several common questions that future passengers flying from New York City airports may have. Note that although the purpose was to help travelers to have a better understanding of NYC flights delay situation, these conclusions should be interpreted as **approximations based on historical data** rather than guarantees for any individual flight.

Q&A to real class presentation questions:

Some suggestions and concerns are collected from the audience of the presentation.

Here are some questions being asked and their answer:

Q: What is the likelihood that my flight will be delayed?

A: Approximately **21% of flights departing from NYC experienced a departure delay of more than 15 minutes**, with arrival delay rates showing a similar pattern. However, most delays are not severe. For travelers seeking transfer, more than 1 hour between flights is suggested to reserve for reducing problems caused by delay.

Q: Does the airport my plane is taking off from affecting the delay rate?

A: An ANOVA analysis confirmed that departure delay rates differ significantly by airport, indicating that airport choice plays a meaningful role in delay likelihood rather than observed differences being due to random variation alone. Based on the delay rate summary table and bar plots, **LaGuardia Airport (LGA)** consistently exhibited the **lowest delay rates**, while **Newark Liberty International Airport (EWR)** showed the **highest departure delay rates**, with **JFK falling between the two**.

However, it is important to note that **statistical significance does not necessarily imply practical significance**. A difference of about **2 percentage points** in delay rates, while statistically detectable, may not be substantial enough to meaningfully affect every traveler's experience.

Q: Does the time period when the flight is scheduled to depart?

A: Time-of-day patterns revealed much stronger differences than airport choice.

Morning flights had the lowest delay rate, at approximately **10%**, making early departures the most reliable option for travelers. In contrast, **night flights experienced the highest delay rate**, exceeding **50%**, likely reflecting the accumulation of delays throughout the day and operational constraints later in the evening. Afternoon and evening flights showed moderate delay rates, with delays generally increasing as the day progresses.

Q: Can you give me some quick suggestions for selecting a flight that is relatively safe from the risk of delays?

A: Based on the analysis, several practical recommendations emerge:

1. **Choose departures in time early in the day, when possible**, as the overall trend is the latter in the day flights are arranged, the more likely to be delayed.
2. **Flying from LGA may reduce delay risk** compared to JFK or EWR according to their 2023 performance, though the difference is not that large.
3. **Delay risk peaks during summer months**, more specifically, June and July, reserve time or plan for the itinerary after the flight if schedule flexibility allows.

While delays cannot be eliminated entirely, understanding these historical patterns still allows passengers to make more informed travel decisions.

Review of group member:

Ziqi Yan: It's hard to comment on him. Overall, I understand his difficulty, but he really did bad on communicating and being responsible. I won't repeat the long descriptions

that are already written in email. When we can do the call, I think we are pretty productive then, however that just happened once, the night before presentation, and for the rest of time I was always desperately waiting for his reply or work.